

ML2025 Fall Homework Assignment 1

Handwritten Solution

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Tools You Need to Know/Learn

- Rank, Nullity, Vector Space, Inner Product Space
- Determinant, Trace, Inverse Matrix, Adjoint Matrix
- Symmetric Matrix, Orthogonal Matrix, Diagonalization
- Positive Definite, Positive Semidefinite
- Eigenvalue Decomposition, Singular Value Decomposition (SVD)

You are expected to know (or learn through this homework) the definitions and usages of the above concepts, and are encouraged to discuss with TAs during TA hour if you encounter difficulties.

Homework Policy

- The official submission deadline will follow the schedule announced on NTU COOL.
- Homework may be handwritten or typed (e.g. using $\text{\LaTeX}$), but must be submitted in **PDF format**.
- If you discuss the homework with classmates, you should state their student IDs in your submission.
- Plagiarism is strictly prohibited. Serious violations will be dealt with according to NTU regulations.

Problem 1 (Linear Algebra Recap)

In the following problems, we begin by recapping key topics from undergraduate linear algebra. We then proceed to derive several fundamental theorems and lemmas, which will play a central role in the forthcoming lectures. In the following problem, we assume V is an inner product space over $\mathbb{R}$ with $\dim V = n$.

Definition 1.1. (Symmetric transformation) A linear transformation $\mathbf{T} : V \rightarrow V$ is called symmetric if

$$\langle \mathbf{T}\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{T}\mathbf{v} \rangle, \quad \forall \mathbf{u}, \mathbf{v} \in V.$$

If $\mathbf{A}$ is the matrix representation of $\mathbf{T}$ and $V \cong \mathbb{R}^n$, this condition is equivalent to

$$\mathbf{u}^\top \mathbf{A} \mathbf{v} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{v}, \quad \forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

Definition 1.2. (positive semidefinite) A symmetric linear transformation $\mathbf{T} : V \rightarrow V$ is called positive semidefinite if

$$\langle \mathbf{T}\mathbf{u}, \mathbf{u} \rangle \geq 0, \quad \forall \mathbf{u} \in V.$$

Equivalently, for the matrix representation $\mathbf{A}$ of $\mathbf{T}$, this condition is equivalent to $\mathbf{A}$ is symmetric and

$$\mathbf{u}^\top \mathbf{A} \mathbf{u} \geq 0, \quad \forall \mathbf{u} \in \mathbb{R}^n.$$

(a) Prove the following statements are equivalent.

- (i) $\mathbf{A}$ is positive semidefinite.
- (ii) $\mathbf{A}$ is symmetric and all of the eigenvalues of $\mathbf{A}$ are all non-negative.
- (iii) There exists some square matrix $\mathbf{B}$ such that $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$.

(Hint: The symmetric matrix is orthogonally diagonalizable).

Solution:

(i) $\Rightarrow$ (ii)

If $\mathbf{A}$ is positive semidefinite, it is orthogonally diagonalizable. Consider $\{\lambda_i\}_{i=1}^n$ the eigenvalues of $\mathbf{A}$ and the corresponding eigenvectors, $\{\mathbf{v}_i\}_{i=1}^n$

$$\langle \mathbf{A}\mathbf{v}_i, \mathbf{v}_i \rangle = \langle \lambda_i \mathbf{v}_i, \mathbf{v}_i \rangle = \lambda_i \|\mathbf{v}_i\|^2 \Rightarrow \lambda_i = \frac{\langle \mathbf{A}\mathbf{v}_i, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2} \geq 0 \quad \forall i = 1, \dots, n$$

(ii) $\Rightarrow$ (i)

Since the symmetric matrix is orthogonally diagonalizable. Thus, $\forall \mathbf{v} \in V$,

$$\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i \quad \alpha_i \in \mathbb{R} \quad \forall i = 1, \dots, n$$

$$\langle \mathbf{A}\mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{A} \sum_{i=1}^n \alpha_i \mathbf{v}_i, \sum_{i=1}^n \alpha_i \mathbf{v}_i \rangle = \langle \sum_{i=1}^n \lambda_i \alpha_i \mathbf{v}_i, \sum_{i=1}^n \alpha_i \mathbf{v}_i \rangle = \sum_{i=1}^n \lambda_i \alpha_i^2 \|\mathbf{v}_i\|^2 \geq 0 \quad (\text{Since } \lambda_i \geq 0)$$

Furthermore, $\mathbf{A}$ is symmetric. Thus $\mathbf{A}$ is positive semidefinite.

(ii) $\Rightarrow$ (iii)

Since $\mathbf{A}$ is symmetric, it is orthogonally diagonalizable.

$$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^\top$$

Where $\mathbf{D}$ is diagonal matrix, and $\mathbf{U}$ is orthogonal matrix. Since the eigenvalues of $\mathbf{A}$ is positive. Let $\sqrt{\mathbf{D}} = \text{diagonal}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})$. Then,

$$\mathbf{A} = \mathbf{U} \sqrt{\mathbf{D}} \sqrt{\mathbf{D}} \mathbf{U}^\top = (\sqrt{\mathbf{D}} \mathbf{U})^\top (\sqrt{\mathbf{D}} \mathbf{U}) = \mathbf{B}^\top \mathbf{B}, \quad \mathbf{B} = \sqrt{\mathbf{D}} \mathbf{U}$$

(iii) $\Rightarrow$ (i)

$$\mathbf{A}^\top = (\mathbf{B}^\top \mathbf{B})^\top = \mathbf{B}^\top \mathbf{B} = \mathbf{A}$$

Therefore, $\mathbf{A}$ is symmetric.

$$\langle \mathbf{A}\mathbf{u}, \mathbf{u} \rangle = \langle \mathbf{B}^\top \mathbf{B}\mathbf{u}, \mathbf{u} \rangle = \langle \mathbf{B}\mathbf{u}, \mathbf{B}\mathbf{u} \rangle = \|\mathbf{B}\mathbf{u}\|^2 \geq 0 \quad \forall \mathbf{u} \in \mathbb{R}^n$$

Hence, $\mathbf{A}$ is positive semidefinite.

In the following section, we will prove an important theorem in machine learning, namely the Singular Value Decomposition (SVD).

Theorem 1 (Singular Value Decomposition). For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, there exist orthogonal matrices $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$, and a diagonal matrix $\Sigma \in \mathbb{R}^{m \times n}$ with nonnegative diagonal entries such that

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top.$$

Consider $\mathbf{A}^\top \mathbf{A}$. It is symmetric and hence orthogonally diagonalizable. Suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthonormal set of eigenvectors of $\mathbf{A}^\top \mathbf{A}$ with $\mathbf{A}^\top \mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i$

(b) Prove that λ_i are nonnegative $\forall i = 1, \dots, n$.

Solution:

$$\begin{aligned} \langle \mathbf{A}\mathbf{v}_i, \mathbf{A}\mathbf{v}_i \rangle &= \langle \mathbf{A}^\top \mathbf{A} \mathbf{v}_i, \mathbf{v}_i \rangle = \lambda_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle = \lambda_i \|\mathbf{v}_i\|^2 \\ &\Rightarrow \lambda_i = \frac{\langle \mathbf{A}\mathbf{v}_i, \mathbf{A}\mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2} \geq 0 \end{aligned}$$

(c) Prove that $\text{Null}(\mathbf{A}^\top \mathbf{A}) = \text{Null}(\mathbf{A})$

Solution:

$$\text{Null}(\mathbf{A}^\top \mathbf{A}) \subseteq \text{Null}(\mathbf{A})$$

$$\begin{aligned} \forall \mathbf{v} \in \text{Null}(\mathbf{A}^\top \mathbf{A}), \quad \langle \mathbf{A}\mathbf{v}, \mathbf{A}\mathbf{v} \rangle &= \langle \mathbf{v}, \mathbf{A}^\top \mathbf{A} \mathbf{v} \rangle = 0 \Rightarrow \|\mathbf{A}\mathbf{v}\|^2 = 0 \\ &\Rightarrow \mathbf{v} \in \text{Null}(\mathbf{A}) \end{aligned}$$

$$\text{Null}(\mathbf{A}^\top \mathbf{A}) \supseteq \text{Null}(\mathbf{A})$$

$$\begin{aligned} \forall \mathbf{v} \in \text{Null}(\mathbf{A}), \quad \mathbf{A}\mathbf{v} &= 0 \Rightarrow \mathbf{A}^\top \mathbf{A} \mathbf{v} = \mathbf{A}^\top 0 = 0 \\ &\Rightarrow \mathbf{v} \in \text{Null}(\mathbf{A}^\top \mathbf{A}) \end{aligned}$$

Suppose $\sigma_i := \sqrt{\lambda_i}$ for each i , and let $r = \text{rank}(\mathbf{A})$. By reordering the set $\{\sigma_1, \dots, \sigma_n\}$, we may assume that

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0.$$

By problem (c), we know $\sigma_r > 0$ and $\sigma_k = 0 \quad \forall k \in \{r+1, \dots, n\}$.

(d) Prove that $\{\mathbf{A}\mathbf{v}_1, \dots, \mathbf{A}\mathbf{v}_r\}$ is an orthogonal set.

Solution:

For $i \neq j$

$$\langle \mathbf{A}\mathbf{v}_i, \mathbf{A}\mathbf{v}_j \rangle = \langle \mathbf{v}_i, \mathbf{A}^\top \mathbf{A} \mathbf{v}_j \rangle = \lambda_j \langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$$

Since $\text{Null}(\mathbf{A}) = \text{Null}(\mathbf{A}^\top \mathbf{A})$. Therefore, $\mathbf{A}\mathbf{v}_i \neq 0, \quad \forall i = 1, \dots, r$.

(e) Let $\mathbf{u}_i = \frac{\mathbf{A}\mathbf{v}_i}{\sigma_i}$, complete the proof the SVD.

(Hint: You may start with $\mathbf{A}(\mathbf{v}_1, \dots, \mathbf{v}_n) = (\sigma_1 \mathbf{u}_1, \dots, \sigma_r \mathbf{u}_r, 0, \dots, 0)$ and using Gram-Schmidt process. Don't forget to show that $\mathbf{U}, \mathbf{V}$ are orthogonal matrices.)

Solution:

Since $\mathbf{A}\mathbf{v}_i$ are orthogonal, $\mathbf{u}_i$ are orthogonal.

$$\|\mathbf{u}_i\|^2 = \frac{\langle \mathbf{A}\mathbf{v}_i, \mathbf{A}\mathbf{v}_i \rangle}{\sigma_i^2} = \frac{\lambda_i \langle \mathbf{v}_i, \mathbf{v}_i \rangle}{\sigma_i^2} = 1$$

$$\mathbf{A}(\mathbf{v}_1, \dots, \mathbf{v}_n) = (\mathbf{u}_1, \dots, \mathbf{u}_r, 0, \dots, 0) \begin{pmatrix} \sigma_1 & 0 & \dots & 0 & \dots & 0 \\ 0 & \ddots & \ddots & 0 & \dots & 0 \\ 0 & 0 & \sigma_r & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 \end{pmatrix}$$

Picking the elements in $\text{R}(\mathbf{A})^\perp$. Using Gram-schmidt process, we can build an orthogonal set $\{\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_m\}$ in $\mathbb{R}^m$. Thus,

$$\mathbf{A}(\mathbf{v}_1, \dots, \mathbf{v}_n) = (\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_m) \begin{pmatrix} \sigma_1 & 0 & \dots & 0 & \dots & 0 \\ 0 & \ddots & \ddots & 0 & \dots & 0 \\ 0 & 0 & \sigma_r & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 \end{pmatrix}$$

Since $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthonormal set. Therefore, $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is an orthogonal matrix. Furthermore, $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_m)$ is also an orthogonal matrix since $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ is an orthonormal set. Let $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top.$$

Problem 2 (Definition of Derivative as Linear Operator)

We now introduce the rigorous definition of the derivative in finite-dimensional vector spaces. The key idea is that the derivative at a point is a linear operator that best approximates the change of the function.

General definition (Fréchet derivative). Let V, W be finite-dimensional real inner product spaces, and let $f : V \rightarrow W$. We say that f is differentiable at $\mathbf{x} \in V$ if there exists a linear operator

$$Df(\mathbf{x}) : V \rightarrow W$$

such that

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - Df(\mathbf{x})[\mathbf{h}]\|_W}{\|\mathbf{h}\|_V} = 0,$$

where $\|\cdot\|_V$ and $\|\cdot\|_W$ denote the norms on V and W respectively. The operator $Df(\mathbf{x})$ is called the derivative (or Jacobian operator) of f at $\mathbf{x}$.

Remark (scalar-valued functions). If $W = \mathbb{R}$, then $Df(\mathbf{x}) : V \rightarrow \mathbb{R}$ is a linear transformation. In this case the definition is equivalent to

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) = Df(\mathbf{x})[\mathbf{h}] + o(\|\mathbf{h}\|_V), \quad \|\mathbf{h}\| \rightarrow 0.$$

If there exists a vector $g \in V$ such that

$$Df(\mathbf{x})[\mathbf{h}] = \langle g, \mathbf{h} \rangle, \quad \forall \mathbf{h} \in V,$$

we call this vector g the gradient of f at $\mathbf{x}$, denoted $\nabla f(\mathbf{x})$. It turns out that a fundamental result in mathematics (the Riesz representation theorem) guarantees that such a vector always exists and is unique (you don't need to know the details in this course, but you can look it up if you're interested).

Coordinate form.

- If $V = \mathbb{R}^n$ with the standard Euclidean inner product, then

$$Df(\mathbf{x}) = \nabla f(\mathbf{x})^\top d\mathbf{x}.$$

- If $V = \mathbb{R}^{m \times n}$ with the Frobenius inner product $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{A}^\top \mathbf{B})$, then

$$Df(\mathbf{X}) = \langle \nabla f(\mathbf{X}), d\mathbf{X} \rangle_F = \text{tr}(\nabla f(\mathbf{X})^\top d\mathbf{X}).$$

Chain rule. If $f : V \rightarrow W$ and $g : U \rightarrow V$ are differentiable at $\mathbf{x} \in U$, then their composition $h = f \circ g : U \rightarrow W$ is differentiable at $\mathbf{x}$ and satisfies

$$Dh(\mathbf{x}) = Df(g(\mathbf{x})) \circ Dg(\mathbf{x}).$$

In the scalar-valued case $f : V \rightarrow \mathbb{R}$, this implies the familiar formula

$$\nabla(f \circ g)(\mathbf{x}) = (\nabla g(\mathbf{x}))^\top \nabla f(g(\mathbf{x})).$$

All the definitions above are just to make sure our calculus is mathematically solid. For the following problems, please use this rigorous way of thinking — that is, start from the small-increment approximation when writing your proofs.

- (a) Given $\mathbf{x}, \mathbf{a} \in \mathbb{R}^n$, show that

$$\nabla \|\mathbf{x} - \mathbf{a}\|_2 = \frac{\partial \|\mathbf{x} - \mathbf{a}\|_2}{\partial \mathbf{x}} = \frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|_2}.$$

Solution:

Let $f(\mathbf{x}) = \|\mathbf{x} - \mathbf{a}\|_2 = \sqrt{(\mathbf{x} - \mathbf{a})^\top (\mathbf{x} - \mathbf{a})}$. We can write $f(\mathbf{x}) = \phi(g(\mathbf{x}))$ where

$$g(\mathbf{x}) = (\mathbf{x} - \mathbf{a})^\top (\mathbf{x} - \mathbf{a}), \quad \phi(t) = \sqrt{t}.$$

Step 1. Gradient of $g(\mathbf{x})$:

$$g(\mathbf{x}) = (\mathbf{x} - \mathbf{a})^\top (\mathbf{x} - \mathbf{a}).$$

For an increment $d\mathbf{x}$,

$$g(\mathbf{x} + d\mathbf{x}) - g(\mathbf{x}) = (\mathbf{x} - \mathbf{a} + d\mathbf{x})^\top (\mathbf{x} - \mathbf{a} + d\mathbf{x}) - (\mathbf{x} - \mathbf{a})^\top (\mathbf{x} - \mathbf{a}) = 2(\mathbf{x} - \mathbf{a})^\top d\mathbf{x} + \|d\mathbf{x}\|^2.$$

So

$$Dg(\mathbf{x})[d\mathbf{x}] = \langle 2(\mathbf{x} - \mathbf{a}), d\mathbf{x} \rangle, \quad \nabla g(\mathbf{x}) = 2(\mathbf{x} - \mathbf{a}).$$

Step 2. Apply chain rule: Since $f(\mathbf{x}) = \phi(g(\mathbf{x}))$,

$$Df(\mathbf{x})[d\mathbf{x}] = \phi'(g(\mathbf{x})) Dg(\mathbf{x})[d\mathbf{x}].$$

Here $\phi'(t) = \frac{1}{2}t^{-1/2}$, so

$$Df(\mathbf{x})[d\mathbf{x}] = \frac{1}{2\sqrt{g(\mathbf{x})}} \cdot \langle 2(\mathbf{x} - \mathbf{a}), d\mathbf{x} \rangle = \left\langle \frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|_2}, d\mathbf{x} \right\rangle.$$

Hence,

$$\nabla \|\mathbf{x} - \mathbf{a}\|_2 = \frac{\mathbf{x} - \mathbf{a}}{\|\mathbf{x} - \mathbf{a}\|_2}, \quad \mathbf{x} \neq \mathbf{a}.$$

(b) Given $\mathbf{a} \in \mathbb{R}^m$, $\mathbf{X} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^n$, show that

$$\nabla(\mathbf{a}^\top \mathbf{X} \mathbf{b}) = \frac{\partial \mathbf{a}^\top \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} = \mathbf{a} \mathbf{b}^\top.$$

Solution:

Let $g(\mathbf{X}) = \mathbf{a}^\top \mathbf{X} \mathbf{b}$ (scalar). Then

$$g(\mathbf{X} + d\mathbf{X}) - g(\mathbf{X}) = \mathbf{a}^\top (d\mathbf{X}) \mathbf{b} = \text{tr}((\mathbf{a} \mathbf{b}^\top)^\top d\mathbf{X}).$$

Hence $Dg(\mathbf{X})[d\mathbf{X}] = \langle \mathbf{a} \mathbf{b}^\top, d\mathbf{X} \rangle_F$ and

$$\nabla(\mathbf{a}^\top \mathbf{X} \mathbf{b}) = \frac{\partial \mathbf{a}^\top \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} = \mathbf{a} \mathbf{b}^\top.$$

(c) Given $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$, show that

$$\nabla(\mathbf{x}^\top \mathbf{A} \mathbf{x}) = \frac{\partial(\mathbf{x}^\top \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}.$$

Solution:

Let $h(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$. Then

$$\begin{aligned} h(\mathbf{x} + d\mathbf{x}) - h(\mathbf{x}) &= (\mathbf{x} + d\mathbf{x})^\top \mathbf{A} (\mathbf{x} + d\mathbf{x}) - \mathbf{x}^\top \mathbf{A} \mathbf{x} \\ &= \mathbf{x}^\top \mathbf{A} d\mathbf{x} + (d\mathbf{x})^\top \mathbf{A} \mathbf{x} + (d\mathbf{x})^\top \mathbf{A} d\mathbf{x} \end{aligned}$$

Note that using the trace and the Frobenius norm $\|\cdot\|_F$, we have

$$(d\mathbf{x})^\top \mathbf{A} d\mathbf{x} = \text{tr}(\mathbf{A} d\mathbf{x} d\mathbf{x}^\top) \leq \|\mathbf{A}\|_F \|d\mathbf{x} d\mathbf{x}^\top\|_F = \|\mathbf{A}\|_F \|d\mathbf{x}\|^2,$$

which implies $(d\mathbf{x})^\top \mathbf{A} d\mathbf{x} = o(\|d\mathbf{x}\|)$ as $d\mathbf{x} \rightarrow 0$. Thus $Dh(\mathbf{x})[d\mathbf{x}] = \langle (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}, d\mathbf{x} \rangle$ and

$$\nabla(\mathbf{x}^\top \mathbf{A} \mathbf{x}) = \frac{\partial(\mathbf{x}^\top \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}. \quad (\text{If } \mathbf{A} \text{ symmetric: } 2\mathbf{A}\mathbf{x}.)$$

Problem 3 (Matrix Calculus)

In this problem, you need to compute the derivative of a scalar function with respect to a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$.

We define

$$\nabla f(\mathbf{A}) = \frac{\partial f(\mathbf{A})}{\partial \mathbf{A}} = \begin{bmatrix} \frac{\partial f}{\partial a_{11}} & \cdots & \frac{\partial f}{\partial a_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial a_{n1}} & \cdots & \frac{\partial f}{\partial a_{nn}} \end{bmatrix}^T = \begin{bmatrix} \frac{\partial f}{\partial a_{11}} & \cdots & \frac{\partial f}{\partial a_{n1}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial a_{1n}} & \cdots & \frac{\partial f}{\partial a_{nn}} \end{bmatrix}$$

(a) Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. Show that

$$\frac{\partial \det(\mathbf{A})}{\partial \mathbf{A}} = \det(\mathbf{A}) (\mathbf{A}^{-1})^T.$$

Hint: Recall from high school that for a 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

where the matrix on the right (before dividing by $\det(\mathbf{A})$) is constructed by taking cofactors and then transposing — this is the adjugate matrix. Similarly, for 3×3 matrices, you may have seen the cofactor expansion formula for the determinant and the adjugate-based formula for the inverse.

In general, for an $n \times n$ matrix $\mathbf{A}$, the cofactor matrix is

$$\mathbf{C} = \begin{bmatrix} C_{11} & \cdots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{n1} & \cdots & C_{nn} \end{bmatrix}, \quad C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the determinant of the $(n-1) \times (n-1)$ minor (delete row i , column j). The transpose of $\mathbf{C}$ is called the adjugate matrix:

$$\text{adj}(\mathbf{A}) = \mathbf{C}^T.$$

One always has the identity

$$\mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}.$$

Try differentiating this identity to obtain the derivative formula.

Solution:

Recall that

$$\det(\mathbf{X}) = \sum_{i=1}^m (-1)^{i+j} x_{ij} M_{ij} = \sum_{i=1}^m x_{ij} C_{ij}$$

where $M_{ij} = \det((x_{mn})_{m \neq i, n \neq j})$ and $C_{ij} = (-1)^{i+j} M_{ij}$.

$$\frac{\partial \det(\mathbf{X})}{\partial x_{ij}} = C_{ij}.$$

Express the derivative as the matrix form

$$\frac{\partial \det(\mathbf{X})}{\partial \mathbf{X}} = \mathbf{C} = \text{adj}(\mathbf{X})^T$$

where

$$\mathbf{C} = \begin{bmatrix} C_{11} & \cdots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{n1} & \cdots & C_{nn} \end{bmatrix}$$

is the cofactor matrix. Thus,

$$\frac{\partial \det(\mathbf{X})}{\partial \mathbf{X}} = \mathbf{C} = \text{adj}(\mathbf{X})^\top = \det(\mathbf{X}) \frac{\text{adj}(\mathbf{X})^\top}{\det(\mathbf{X})} = \det(\mathbf{X}) (\mathbf{X}^{-1})^\top.$$

(b) Prove that

$$\frac{\partial \log(\det(\mathbf{A}))}{\partial a_{ij}} = \mathbf{e}_j^\top \mathbf{A}^{-1} \mathbf{e}_i,$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a (non-singular) matrix, and $\mathbf{e}_j$ is the unit vector along the j -th axis (e.g. $\mathbf{e}_3 = [0, 0, 1, 0, \dots, 0]^\top$). It is common to write the formula as

$$\frac{\partial \log(\det(\mathbf{A}))}{\partial \mathbf{A}} = (\mathbf{A}^{-1})^\top.$$

Hint: You can continue directly from part (a), apply the chain rule to $\log(\det(\mathbf{A}))$.

Alternative hint (prove (b) $\Rightarrow$ (a)). You may also start from Cramer's rule: for $\mathbf{A}\mathbf{x} = \mathbf{e}_i$,

$$x_j = \frac{\det(\mathbf{A}_j)}{\det(\mathbf{A})},$$

where $\mathbf{A}_j$ is obtained from $\mathbf{A}$ by replacing its j -th column with $\mathbf{e}_i$. By differentiating $\log \det(\mathbf{A})$ along coordinate directions and applying Cramer's rule, you can derive part (b). If you proceed this way and then recover part (a) from (b) via the chain rule, you will still receive full credit.

Solution:

Suppose that $\mathbf{A}$ is invertible. Observe that

$$\frac{\partial \log(\det(\mathbf{A}))}{\partial a_{ij}} = \frac{1}{\det(\mathbf{A})} \frac{\partial \det(\mathbf{A})}{\partial a_{ij}}$$

Recall that for any square matrix $\mathbf{A}$ and fixed $1 \leq j \leq m$,

$$\det(\mathbf{A}) = \sum_{i=1}^m (-1)^{i+j} a_{ij} M_{ij} = \sum_{i=1}^m a_{ij} C_{ij}$$

where $M_{ij} = \det(a_{mn})_{m \neq i, n \neq j}$ and $C_{ij} = (-1)^{i+j} M_{ij}$. Remember the relation

$$\mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}.$$

Fix $1 \leq i, j \leq m$ and expand $\det(\mathbf{A}) = \sum_{k=1}^m a_{kj} C_{kj}$. Then

$$\begin{aligned} &= \frac{1}{\det(\mathbf{A})} \frac{\partial \det(\mathbf{A})}{\partial a_{ij}} \\ &= \frac{1}{\det(\mathbf{A})} \frac{1}{\partial a_{ij}} \sum_{k=1}^m a_{kj} C_{kj} \\ &= \frac{1}{\det(\mathbf{A})} C_{ij} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})_{ji}. \end{aligned}$$

Since $\text{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{A}^{-1}$, we have

$$\frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})_{ji} = (\mathbf{A}^{-1})_{ji}.$$

Thus,

$$\frac{\partial \log(\det(\mathbf{A}))}{\partial \mathbf{A}} = (\mathbf{A}^{-1})^\top.$$

Problem 4 (Closed-Form Linear Regression Solution) (20 + 15 pts)

Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon},$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\boldsymbol{\theta} \in \mathbb{R}^d$ and $\boldsymbol{\epsilon} \in \mathbb{R}^n$. Denote $\mathbf{X}_i \in \mathbb{R}^{1 \times d}$ as the i -th row of $\mathbf{X}$, with the following interpretations:

- If the linear model has the bias term, then write $\boldsymbol{\theta} = [w_1, \dots, w_m, b]^\top$ and $\mathbf{X}_i = [x_{i,1}, x_{i,2}, \dots, x_{i,m}, 1]$, namely $d = m + 1$.
- If the linear model has no bias term, then write $\boldsymbol{\theta} = [w_1, \dots, w_d]^\top$ and $\mathbf{X}_i = [x_{i,1}, x_{i,2}, \dots, x_{i,m}]$, namely $d = m$.

(a) (20 pts) Without the bias term, consider the L^2 -regularized loss function:

$$\sum_i \kappa_i (y_i - \mathbf{X}_i \boldsymbol{\theta})^2 + \lambda \sum_j w_j^2, \quad \lambda > 0, \kappa_i > 0 \text{ for all } i.$$

Show that the optimal solution that minimizes the loss function is $\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{K} \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{K} \mathbf{y}$, where

$$\mathbf{K} = \begin{bmatrix} \kappa_1 & & 0 \\ & \ddots & \\ 0 & & \kappa_n \end{bmatrix}$$

is a diagonal matrix and $\mathbf{I}$ is the $d \times d$ identical matrix.

Solution:

First, represent the loss function as

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \mathbf{K} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) + \lambda \boldsymbol{\theta}^\top \boldsymbol{\theta}$$

Next, take gradient of $\boldsymbol{\theta}$ and set it to 0, you will get the optimal solution $\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{K} \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{K} \mathbf{y}$

(b) (Bonus) (15 pts) With the bias term, the L^2 -regularized loss function becomes

$$\sum_i \kappa_i (y_i - \mathbf{X}_i \boldsymbol{\theta})^2 + \lambda \sum_j w_j^2, \quad \lambda > 0, \kappa_i > 0 \text{ for all } i.$$

Show that the optimal solution that minimizes the loss function is $\boldsymbol{\theta}^* = [\mathbf{w}^*, b^*]^\top$, where

$$\begin{aligned} \mathbf{w}^* &= \left(\tilde{\mathbf{X}}^\top \mathbf{K} \tilde{\mathbf{X}} + \lambda \mathbf{I} - \frac{1}{\text{Tr}(\mathbf{K})} \tilde{\mathbf{X}}^\top \mathbf{K} \mathbf{e} \mathbf{e}^\top \mathbf{K} \tilde{\mathbf{X}} \right)^{-1} \tilde{\mathbf{X}}^\top \mathbf{K} \left(\mathbf{y} - \frac{1}{\text{Tr}(\mathbf{K})} \mathbf{e} \mathbf{e}^\top \mathbf{K} \mathbf{y} \right), \\ b^* &= \frac{1}{\text{Tr}(\mathbf{K})} (\mathbf{e}^\top \mathbf{K} \mathbf{y} - \mathbf{e}^\top \mathbf{K} \tilde{\mathbf{X}} \mathbf{w}^*) \end{aligned}$$

for which $\mathbf{e} = [1 \dots 1]^\top$ denotes the all one vector, $\mathbf{X} = [\tilde{\mathbf{X}} \mathbf{e}]$, where $\tilde{\mathbf{X}}$ is the dataset of the no bias term. $\text{Tr}(\mathbf{K})$ is the trace of the matrix $\mathbf{K}$, and that $\mathbf{K}$ and $\mathbf{I}$ are defined as in (a).

Solution:

First, represent the loss function as

$$(\mathbf{y} - \tilde{\mathbf{X}} \mathbf{w} - b \mathbf{e})^\top \mathbf{K} (\mathbf{y} - \tilde{\mathbf{X}} \mathbf{w} - b \mathbf{e}) + \lambda \mathbf{w}^\top \mathbf{w}$$

Next, take gradient of both $\mathbf{w}$ and b and set them to 0 respectively, you will get two equations. By solving the system of equations carefully, you will get the optimal solution $\boldsymbol{\theta}^* = [\mathbf{w}^{*T}, b^*]^T$, where

$$\begin{aligned}\mathbf{w}^* &= \left(\tilde{\mathbf{X}}^T \mathbf{K} \tilde{\mathbf{X}} + \lambda \mathbf{I} - \frac{1}{\text{Tr}(\mathbf{K})} \tilde{\mathbf{X}}^T \mathbf{K} \mathbf{e} \mathbf{e}^T \mathbf{K} \tilde{\mathbf{X}} \right)^{-1} \tilde{\mathbf{X}}^T \mathbf{K} \left(\mathbf{y} - \frac{1}{\text{Tr}(\mathbf{K})} \mathbf{e} \mathbf{e}^T \mathbf{K} \mathbf{y} \right), \\ b^* &= \frac{1}{\text{Tr}(\mathbf{K})} \left(\mathbf{e}^T \mathbf{K} \mathbf{y} - \mathbf{e}^T \mathbf{K} \tilde{\mathbf{X}} \mathbf{w}^* \right)\end{aligned}$$

Problem 5 (Noise and Regularization) (20 pts)

Consider the linear model $f_{\mathbf{w},b} : \mathbb{R}^k \rightarrow \mathbb{R}$, where $\mathbf{w} \in \mathbb{R}^k$ and $b \in \mathbb{R}$, defined as

$$f_{\mathbf{w},b}(x) = \mathbf{w}^T \mathbf{x} + b$$

Given dataset $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, if the inputs $\mathbf{x}_i \in \mathbb{R}^k$ are contaminated with input noise $\boldsymbol{\eta}_i \in \mathbb{R}^k$, we may consider the expected sum-of-squares loss in the presence of input noise as

$$\tilde{L}_{ss}(\mathbf{w}, b) = \mathbb{E} \left[\frac{1}{2N} \sum_{i=1}^N (f_{\mathbf{w},b}(\mathbf{x}_i + \boldsymbol{\eta}_i) - y_i)^2 \right]$$

where the expectation is taken over the randomness of input noises $\boldsymbol{\eta}_1, \dots, \boldsymbol{\eta}_N$. Additionally, the inputs $(\mathbf{x}_i)$ and the input noise $(\boldsymbol{\eta}_i)$ are independent.

Now assume the input noises $\boldsymbol{\eta}_i = [\eta_{i,1}, \eta_{i,2}, \dots, \eta_{i,k}]^T$ are random vectors with zero mean $\mathbb{E}[\eta_{i,j}] = 0$, and the covariance between components is given by

$$\mathbb{E}[\eta_{i,j} \eta_{i',j'}] = \delta_{i,i'} \delta_{j,j'} \sigma^2$$

where $\delta_{i,i'} = \begin{cases} 1 & , \text{ if } i = i' \\ 0 & , \text{ otherwise.} \end{cases}$ denotes the Kronecker delta.

(a) (20 pts) Please show that

$$\tilde{L}_{ss}(\mathbf{w}, b) = \frac{1}{2N} \sum_{i=1}^N (f_{\mathbf{w},b}(\mathbf{x}_i) - y_i)^2 + \frac{\sigma^2}{2} \|\mathbf{w}\|^2$$

That is, minimizing the expected sum-of-squares loss in the presence of input noise is equivalent to minimizing noise-free sum-of-squares loss with the addition of a L^2 -regularization term on the weights.

(Hint: $\|\mathbf{x}\|^2 = \mathbf{x}^T \mathbf{x} = \text{tr}(\mathbf{x} \mathbf{x}^T)$ and the square of a vector is dot product with itself)

Solution:

By definition,

$$\begin{aligned} \tilde{L}_{ss}(\mathbf{w}, b) &= \mathbb{E} \left[\frac{1}{2N} \sum_{i=1}^N (f_{\mathbf{w},b}(\mathbf{x}_i + \boldsymbol{\eta}_i) - y_i)^2 \right] \\ &= \frac{1}{2N} \sum_{i=1}^N \mathbb{E} \{ (\mathbf{w}^T (\mathbf{x}_i + \boldsymbol{\eta}_i) - y_i)^2 \} \\ &= \frac{1}{2N} \sum_{i=1}^N \mathbb{E} [\{ (\mathbf{w}^T \mathbf{x}_i - y_i) + \mathbf{w}^T \boldsymbol{\eta}_i \}^2] \\ &= \frac{1}{2N} \sum_{i=1}^N \mathbb{E} [(\mathbf{w}^T \mathbf{x}_i - y_i)^2] - 2 \mathbb{E} \{ \mathbf{w}^T \boldsymbol{\eta}_i (\mathbf{w}^T \mathbf{x}_i - y_i) \} + \mathbb{E} [(\mathbf{w}^T \boldsymbol{\eta}_i)^2] \\ &= \frac{1}{2N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2 - 2 \mathbf{w}^T (\mathbf{w}^T \mathbf{x}_i - y_i) \mathbb{E}(\boldsymbol{\eta}_i) + \mathbb{E} [(\mathbf{w}^T \boldsymbol{\eta}_i)^2] \\ &= \frac{1}{2N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2 + \mathbb{E} [(\mathbf{w}^T \boldsymbol{\eta}_i)^2] \end{aligned}$$

Note that $\mathbb{E}(\eta_i) = 0$ Now, calculate $\mathbb{E}[(\mathbf{w}^T \eta_i)^2]$

$$\begin{aligned}
\sum_{i=1}^N \mathbb{E}(\mathbf{w}^T \eta_i)^2 &= \sum_{i=1}^N \mathbb{E}\left(\sum_{j=1}^k w_j \eta_{i,j}\right)^2 \\
&= \sum_{i=1}^N \mathbb{E}\left(\sum_{j=1}^k \sum_{l=1}^k w_j w_l \eta_{i,j} \eta_{i,l}\right) \\
&= \sum_{j=1}^k \sum_{l=1}^k w_j w_l \sum_{i=1}^N \mathbb{E}(\eta_{i,j} \eta_{i,l}) \\
&= N \sigma^2 \sum_{j=1}^k \sum_{l=1}^k w_j w_l = N \sigma^2 \|\mathbf{w}\|^2
\end{aligned}$$

Hence,

$$\begin{aligned}
\tilde{L}_{ss}(\mathbf{w}, b) &= \frac{1}{2N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2 + \frac{1}{2N} N \sigma^2 \|\mathbf{w}\|^2 \\
&= \frac{1}{2N} \sum_{i=1}^N (f_{\mathbf{w},b}(\mathbf{x}_i) - y_i)^2 + \frac{\sigma^2}{2} \|\mathbf{w}\|^2
\end{aligned}$$